Kimberly L Ray, PhD

Research Scientist
Department of Psychology, University of Texas at Austin

Contact Information:
Email: kimberlylray@gmail.com;
kimray@utexas.edu

Kimberly L. Ray, PhD, is a neuroscientist and Research Scientist in the Department of Psychology at The University of Texas at Austin. Her research examines how the brain's functional networks and white matter connections support cognition and how these systems differ across health and clinical conditions. She combines functional and structural neuroimaging, electroencephalography, and neuroimaging meta-analysis to study brain organization, with work spanning cognitive control, memory, depression, and schizophrenia.

Alongside her empirical research, Dr. Ray works on research governance, scientific infrastructure, and international collaboration. Her experience includes preparing and revising institutional review board (IRB) materials for human-subjects research, overseeing institutional Data Use Agreements (DUAs), and supporting controlled-data access and sharing workflows. As Project Manager for the NIH BRAIN Initiative's BRAIN CONNECTS APEX Data Coordinating Center, she manages budgets, schedules, and milestones across scientific and technical teams. She previously directed delivery of the Wellcome-funded BRIDGE international data governance program.

Dr. Ray also serves as a maintainer within the Brain Imaging Data Structure (BIDS) community, contributing to and co-leading extensions for connectivity derivatives and brain atlases. She leads the Data Governance Theme of the Neuroscience Interest Group, a collaboration involving BRIDGE, the International Neuroinformatics Coordinating Facility (INCF), and the Global Alliance for Genomics and Health (GA4GH). Through these roles and her contributions to the brainlife.io cloud computing platform, she advances community standards, reproducible research, and resources for responsible brain-data sharing across institutions and national borders.

Her current interests include linking structural connectivity with functional brain networks, developing AI-enabled approaches to neuroimaging evidence synthesis, and advancing international frameworks for responsible data sharing. Through teaching, workshops, and mentorship, she also supports students and researchers in neuroimaging analysis, open science, and collaborative research.

EDUCATION

2010-2014 Ph.D.
University of Texas Health Science Center San Antonio, San Antonio, TX
Graduate School of Biomedical Sciences, Radiological Sciences
Concentration: Neuroscience Imaging Advisor: Angela R. Laird, PhD

2005-2009 B.A. *Magna Cum Laude*
Texas Lutheran University, Seguin, TX
Major/Minor: Physics/Math Graduated in the Honors Program

International Education

2008 Central College Abroad, London, UK
London Metropolitan University, London, UK

Employment

At The University of Texas

2026 – present	Research Scientist	Department of Psychology
2019 – 2026	Research Assistant Professor,	Department of Psychology
2017 – 2019	Post Doctoral Fellow,	Department of Psychology
	Supervisor: David Schnyer, PhD.	

Prior to The University of Texas

2014 – 2017	Post-doctoral Scholar,	Department of Psychiatry and Behavioral Sciences University of California, Davis.
	Supervisor: Cameron Carter, MD.	

PUBLICATIONS

(*Google Scholar; h-index: 23, over 5,000 citations total*)

Peer-Reviewed

1) Botes M, Lavazza A and **Ray KL** (2026) Editorial: Legal and ethical reforms for neurotechnological advancements in the global south: perspective and challenges. Front. Hum. Neurosci. 20:1985765. doi: 10.3389/fnhum.2026.1985765
2) **Ray KL**, M Botes, M Collier, A Yi, S Hershman, T Monteiro, M Genuino, JP Poline, D Eke, R Chavarriaga, AO Ihunwo, FT Moll, T Nichols, F Pestilli. Global brain research in the era of national data sovereignty. Nature Neuroscience. 2026. https://doi.org/10.1038/s41593-026-02468-w
3) Hanekamp S, **Ray KL**, Caron B, Thavornfung J, Flexwala A, Mehboob MF, and Pestilli F. Lifespan lateralization of human brain white matter. *Communications Biology. In Press (2026).*
4) Chen K, Lacey JN, Sanguinetti JL, **Ray KL**, Schney DM, Allen JJB. Simultaneous EEG and fMRI investigation of frontal alpha power and frontal alpha asymmetry in young adults with a history of major depressive disorder. Journal of Affective Disorders, Volume 398, 2026,120905, ISSN 0165-0327, doi.org/10.1016/j.jad.2025.120905.
5) Delogu F, Aspinal C, **Ray KL**, Solon Heinsfeld A, Victory C, Pestilli F. Breaking barriers: broadening neuroscience education via cloud platforms and course-based undergraduate research. Frontiers in Neuroinformatics, 2025
6) Hayashi S, Caron BA,.. **Ray KL**, MacKenzie A, … Pestilli F. brainlife.io: A decentralized and open source cloud platform to support neuroimaging research. *Nat Methods **21**, 809–813 (2024). https://doi.org/10.1038/s41592-024-02237-2*
7) Peraza JA, Salo T, Riedel MC, Bottenhorn KL, Poline JB, Dockès J, Kent JD, Bartley JE, Flannery JS, Hill-Bowen L, Pintos Lobo R, Poudel R, **Ray KL**, Robinson JL, Laird RW, Sutherland MT, de la Vega A, Laird AR. Meta-analytic decoding of the cortical gradient of functional connectivity. Imaging Neuroscience, 2 (1-32), 2024. doi: https://doi.org/10.1101/2023.08.01.551505
8) **Ray KL**, Griffin NR, Shumake J, Alario A, Allen AAJ, Beevers CG, Schnyer DM. Altered electroencephalography resting state network coherence in remitted MDD. Brain Research, 1806 (1) May 2023.
9) Ramage AE, **Ray KL**, Franz HM, Tate DF, Lewis JD, Robin D. Cingulo-Opercular and Frontoparietal Network Control of Effort and Fatigue in Mild Traumatic Brain Injury. Frontiers in Human Neuroscience, Cognitive Neuroscience.
10) Don H, Davis T, **Ray KL**, McMahon M, Cornwall A, Schnyer DM, Worthy DA. Neural representations of gain-loss frequency in older and younger adults. Neurobiology of Aging, 109, 247-258, 2022.

11) Meier SK, **Ray KL**, Mastan JC, Salvage SR, Robin DA. Meta-analytic connectivity modelling of deception-related brain regions. 2021, Accepted at PLOS ONE.
12) Aytur SA, **Ray KL**, Meier SK, Campbell J, Gendron BC, Waller NC, Robin DA. Neural mechanisms of acceptance and commitment therapy for chronic pain: a network-based fMRI approach. Frontiers in human neuroscience 15, 28, 2021.
13) Meier SK, **Ray KL**, Waller NC, Gendron BC, Aytur SA, Robin DA. Network analysis of induced neural plasticity post-acceptance and commitment therapy for chronic pain. Brain Sciences 11 (1), 10, 2021.
14) **Ray KL**, Ragland JD, MacDonald A, Gold J, Silverstein S, Barch D, Carter CS. Dynamic Reorganization of the Frontal Parietal Network During Cognitive Control and Episodic Memory. CABN, 1-15, 2019.
15) Hsu KJ, Caffey K, Pisner D, Shumake J, Risom S, **Ray KL**, Smits JAJ, Schnyer DM, Beevers CG. Attentional bias modification treatment for depression: Study protocol for a randomized controlled trial. Contemp Clin Trials. 2018 Dec;75:59-66. doi: 10.1016/j.cct.2018.10.014.
16) Niendam T, **Ray KL**, Iosif AM, Lesh TA, Ashby SR, Patel PK, Smucny J, Ferrer E, Solomon M, Ragland JD, Carter CS. Association of Age at Onset and Longitudinal Course of Prefrontal Function in Youth with Schizophrenia. JAMA psychiatry, 2018. doi:10.1001/jamapsychiatry.2018.2538
17) Riedel MC, Yanes JA, **Ray KL**, Eickhoff SB, Fox PT, Sutherland MT, Laird AR. Dissociable meta-analytically identified brain networks contribute to coordinated emotional processing. Human Brain Mapping 39 (6), 2514-2531, 2018.
18) Yanes JA, Riedel MC, **Ray KL**, Kirkland A, Bird R, Boeving E, Reid M, Gonzalez R, Robinson J, Laird AR, Sutherland, M. Neuroimaging Meta-Analysis of Cannabis-Use Studies Reveals Convergent Functional Alterations in Brain Regions Supporting Cognitive Control and Reward Processing. Journal of Psychopharmacology, 32 (3), 283-295, 2018.
19) **Ray KL,** Lesh T, Howell A, Salo T, Ragland JD, MacDonald A, Gold J, Silverstein S, Barch D, Carter CS. Functional Network Changes and Cognitive Control in Schizophrenia. NeuroImage: Clinical 15, 161-170, 2017.
20) Laird AR, Riedel MC, Okoe M, Jianu R, **Ray KL**, Eickhoff SB, Smith MS, Fox PT, Sutherland MT. Heterogeneous fractionation profiles of meta-analytic coactivation networks. NeuroImage, 149, 424-435, 2017.
21) **Ray KL**, Zald DH, Bludau S, Riedel MC, Bzdok D, Yanes J, Falcone KE, Amunts K, Fox PT, Eickhoff SB, Laird AR. Connectivity-based parcellation of the frontal pole. NeuroImage, 123, 200-211, 2015.
22) Riedel MC, **Ray KL**, Dick AS, Hernandez Z, Fox PM, Eickhoff SB, Fox PT, Laird AR. Intrinsic connectivity and behavioral parcellation of the human cerebellum through meta-analytic modeling. NeuroImage, 117, 327-342, 2015.
23) Laird AR, Riedel MC, Sutherland MT, Eickhoff SB, **Ray KL**, Uecker AM, Fox PM, Turner JA, Fox PT. Neural architecture underlying classification of face perception paradigms. NeuroImage, 119, 70-80, 2015.
24) Sutherland MT, **Ray KL**, Riedel MC, Yanes JA, Stein EA, Laird AR. Neurofunctional impact of nicotinic acetylcholine receptor stimulation: An ALE meta-analysis of pharmacological neuroimaging studies. Biological Psychiatry, 78(10), 711-720, 2015.
25) Acheson A, **Ray KL**, Li K, Fox PT, Lovallo WR, Laird AR. Functional activation and effective connectivity differences in adolescent marijuana users performing a simulated gambling task. Journal of Addiction, 2015. http://dx.doi.org/10.1155/2015/783106

26) Flagmeier SG, **Ray KL**, Parkinson AL, Li K, Vargas R, Laird AR, Larson CR, Robin DA. The critical role of superior temporal gyrus in voice control. Brain and Language, 132, 7-13, 2014.

27) **Ray KL**, McKay DR, Fox PM, Riedel MC, Eucker AM, Beckmann CF, Smith SM, Fox PT, Laird AR. ICA model order selection of task co-activation networks. Frontiers in Neuroscience, 7, 237, 2013. doi: 10.3389/fnins.2013.00237

28) Zald DH, McHugo M, **Ray KL,** Glahn DC, Eickhoff SB. Laird AR. Meta-analytic connectivity modeling reveals differential functional connectivity of the medial and lateral orbitofrontal cortex. Cereb Cortex, 24(1), 232-248, 2013.

29) Laird AR, Eickhoff SB, Rottschy C, Bzdok D, **Ray KL**, Fox PT. Networks of task co-activations. NeuroImage, 80, 505 – 514, 2013.

30) Niendam T, Laird AR, **Ray KL**, Dean YM, Glahn DC, Carter CS. Meta-analytic evidence for a superordinate cognitive control network subserving diverse executive functions. Cogn Affect Behav Neurosci, 12, 241 – 268, 2012.

31) Laird AR, Eickoff SB, Fox PM, Uecker AM, **Ray KL**, Saenz JJ, McKay DR, Bzdok D, Laird RW, Robinson JL, Turner JA, Turkeltaub PE, Lancaster JL, Fox PT. The BrainMap strategy for standardization, sharing, and meta-analysis of neuroimaging data. BMC Res Notes 4, 349, 2011.

32) Laird AR, Fox PM, Eickhoff SB, Turner JA, **Ray KL**, McKay DR, Glahn DC, Beckmann CF, Smith SM, Fox PT. Behavioral interpretations of intrinsic connectivity networks. J Cogn Neurosci, 23, 4022 – 4037 2011.

33) Laird AR, Robinson JL, McMillan KM, Tordesillas-Gutierrez D, Moran ST, Gonzales Flagmeier SM, **Ray KL**, Franklin C, Glahn DC, Fox PT, Lancaster JL. Comparison of the disparity between Talairach and MNI coordinates in functional neuroimaging data: Validation of the Lancaster transform. Neuroimage 51, 677 –683, 2010.

Pre-prints

1) **Ray KL**, Kim S, Herrera K, Guo S, Reyes D, Pereza JA, Bostoganashvilli A, Amorosino G, Laird AR, Pestilli F. Human white matter tracts aligned with canonical functional brain networks. https://doi.org/10.64898/2026.05.20.726546. *Under review at Nature Communications*.
2) **Ray KL,** Shumake J, Laird AR, Beevers CG, Schnyer DM. Reduced network connectivity in healthy children with maternal history of depression in the ABCD cohort. https://doi.org/10.31234/osf.io/jwcqk

Book Chapters

1) Aytur S, Roy G, Doyle M, **Ray KL**, Meier S, Robin D. Linking fMRI, pain and addictions. Handbook of Substance Misuse and Addictions: From Biology to Public Health. 2021 Accepted, in production at Springer Nature.
2) **Ray KL,** Laird AR**.** Web networks, meta-analysis, and databasing. Encyclopedia of Computational Neuroscience, Article ID: 348607, Chapter ID: 542, Springer Reference, 2013.
3) Riedel MC, **Ray KL**, Fox PM, Uecker A, Eickhoff SB, Fox PT, Laird AR. BrainMap. Encyclopedia of Computational Neuroscience, Article ID: 348551, Chapter ID: 486, Springer Reference, 2013.

Other Publications

1) Malhotra A, Mathews JD, **Ray KL**. Aspect sensitivity considerations in determining day-to-night variations in meteor trail durations. Geophysical Research Letters, 36 (22), 2009.

PEER-REVIEWED ABSTRACTS AND CONFERENCE PROCEEDINGS

1) Global Challenges and Research Capacity in Neuroimaging. Porter B, Collier M, Yi A, F. Troval-Moll, A Ihunwo, R Chavarriaga, T. Nichols, Ray KL, Pestilli F. OHBM 2026.
2) BRIDGES to cross-border neuroimaging data transfers: Legal bottlenecks and enablers from a four-region comparative analysis. Botes M, Collier M, Ray KL, Pestilli F. OHBM 2026.
3) "Meta-analytic functional decoding of the cortical terminations of the 45 major white matter tracts" Kim S, Reyes D, Peraza J, GuoS, Laird A, Ray KL, Pestilli F. OHBM 2025
4) EzGOV: A tool to find, develop, and manage data governance documents. Heinsfeld A, Bhatia D, Lee N, Collier M, Ray KL, Pestilli F. Accepted at the OHBM, SfN, and BRAIN CONNECTS 2025 meetings.
5) "Developing an International Data Governance Framework for Brain Data". Ray KL, A. MacKenzie, D Eke, M. Brown, F. Troval-Moll, A Ihunwo, R Chavarriaga, T. Nichols, and F. Pestilli Accepted at the NIH BRAIN Initiative, INCF, and SfN 2023 meetings.
6) "A probabilistic atlas of white matter tracts and their functional associations". Fernández A, Caron B, Ray KL, Guo S, Thavornfung J, Pestilli F. Organization of Human Brain Mapping: 2023.
7) "Meta-analytic connectivity model for a superordinate cognitive control network". Natal S, Ray KL, Schnyer DM. Organization of Human Brain Mapping: 2022.
8) "Rest activity rhythm stability and functional network interactions during episodic retrieval". McMahon M, Ray KL, Schnyer DM. Organization of Human Brain Mapping: 2022.
9) "Emotion and attention control effects in depression". Jumonville G, Ray KL, Schnyer DM. Organization of Human Brain Mapping: 2022.
10) "Dynamic network coupling associated with effort and fatigue in Traumatic Brain Injury". Ramage AE, Leach J, Franz HM, Ray KL, Robin DA. Organization of Human Brain Mapping: 2022.
11) "Meta-Analytic Connectivity Modelling of Psychopathic Inmate and Non-inmate Brain Networks". Salvage SR, Meier SK, Ray KL, Robin DA. Organization of Human Brain Mapping: 2022.
12) "Altered resting state network coherence in remitted MDD". Ray KL, Shumake J, Beevers C, Schnyer D. Organization of Human Brain Mapping: Virtual, 2021.
13) "Relationship between elevated depressive symptoms, brain volume, and negative attention bias". Baig J, Castillo A, Ray KL, Harle J, McConley I, McMahon M, Hsu K, Shumake J, Beevers C, Schnyer D. Organization of Human Brain Mapping: Virtual, 2021.
14) "Meta-analytic connectivity modelling of deception-related brain regions." Meier SK, Ray KL, Robin DA. Organization of Human Brain Mapping: Virtual, 2021.
15) "Reliability of resting state connectivity networks in individuals with elevated depression severity". Ray KL, Shumake J, Beevers C, Schnyer DM. Organization of Human Brain Mapping: Montreal, 2020.
16) "Dynamic network coupling associated with effort and fatigue in traumatic brain injury". Ramage AE, Ray KL, New AB, Franz HM, Tate DF, Lewis JD, Robin DA. Organization of Human Brain Mapping: Montreal, 2020.
17) "Neural foundations of chronic pain: an ALE meta-analysis of regional brain activation". Waller N, Aytur S, Ray KL, Robin DA. Organization of Human Brain Mapping: Montreal, 2020.

18) "Graph theory analysis of chronic pain patients pre and post Acceptance and Commitment Therapy ". Meier S, Aytur S, Ray KL, Robin, DA. Organization of Human Brain Mapping: Montreal, 2020.
19) "Corresponding Relationships between Resting State EEG power and connectivity". Ray KL, Griffin N, Alario A, Stewart R, Labrada J, Beevers C, Schnyer DM. Organization of Human Brain Mapping: Rome, 2019.
20) "Age related differences in resting state network functional connecitivty and associations with sleep-wake cycles". McMahon MC, Ray KL, Pisner D, Gandy L, Schnyer DM. Organization of Human Brain Mapping: Rome, 2019.
21) "Relationships between Depression Vulnerability and Brain Networks: a multimodal EEG fMRI study". Ray KL, Allen J, Sanguinetti J, Schnyer DM. Organization of Human Brain Mapping: Singapore, 2018.
22) "Dynamic Network Reorganization of the Frontal-Parietal, Salience, and Default Mode Networks during Cognitive Control and Episodic Memory". Ray KL, MacDonald AW, Ragland JD, Gold JM, Silverstein SM, Barch DM, Carter CS. Society of Biological Psychiatry: San Diego, USA, 2017. Organization of Human Brain Mapping: Vancouver, Canada, 2017.
23) "Neurobiological Impacts of Long-Term Cannabis Use: An ALE Meta-Analysis of Neuroimaging Studies". Yanes JA, Riedel MC, Ray KL, Robinson JL, Laird AR, Sutherland MT. Organization of Human Brain Mapping: Geneva, Switzerland, 2016.
24) "Functional network properties of episodic memory". Ray KL, Salo T, Lesh TA, MacDonald AW, Ragland JD, Gold JM, Silverstein SM, Barch DM, Carter CS. Organization of Human Brain Mapping: Honolulu, USA, 2015.
25) "Functional specificity of rest and task based modules: A graph theoretical analysis". Ray KL, Riedel MC, Uecker A, Yanes JA, Fox PT, Laird AR. Organization of Human Brain Mapping: Honolulu, USA, 2015.
26) "Cognitive Control of emotion: A meta-analytic network approach to understanding emotion". Yanes JA, Riedel MC, Ray KL, Fox PT, Sutherland MT, Laird AR. Organization of Human Brain Mapping: Honolulu, USA, 2015.
27) "nAChR agonists induce heterogeneous functional alterations across the brain: An ALE meta-analysis". Sutherland MT, Ray KL, Riedel MC, Yanes JA, Stein EA, Laird AR. Organization of Human Brain Mapping: Honolulu, USA, 2015.
28) "Towards a prediction of cognitive deficits based on underlying connectivity differences in schizophrenia". Ray KL, Damaraju E, Riedel MC, Calhoun VD, Uecker A, Eickhoff SB, Fox PT, Laird AR. Organization of Human Brain Mapping: Hamburg, Germany, 2014.
29) "Transition between parcellation regions within the anterior cingulate: A meta-analytic approach". Riedel MC, Ray KL, Eickhoff SB, Fox PT, Laird AR. Organization of Human Brain Mapping: Hamburg, Germany, 2014.
30) "Representational similarity analysis reveals functional connectivity disruptions in schizophrenia". Riedel MC, Ray KL, Turner JA, Fox PT, Laird AR. Organization of Human Brain Mapping: Hamburg, Germany, 2014.
31) "A Meta-Analysis of Problem-Solving Activations Within Mathematical and Verbal Domains". Bartley JE, Ray KL, Riedel MC, Yanes JA, Fox PT, Brewe E, Laird AR. Organization of Human Brain Mapping: Hamburg, Germany, 2014.

32) "Connectivity-based parcellation of the frontal pole". Ray KL, Bludau S, Riedel MC, Hernandez ZR, Zald DH, Fox PT, Eickhoff SB, Laird AR. Organization of Human Brain Mapping: Seattle, USA, 2013.

33) "Intrinsic connectivity and behavioral parcellation of the cerebellum through meta-analytic modeling". Riedel MC, Ray KL, Hernandez ZR, Fox PM, Uecker A, Eickhoff SB, Fox PT, Laird AR. Organization of Human Brain Mapping: Seattle, USA, 2013.

34) "Data-driven refinement of cognitive paradigm classifications: A face discrimination meta-analysis". Hernandez ZR, Ray KL, Riedel MC, Eickhoff SB, Fox PT, Turner JA, Laird AR. Organization of Human Brain Mapping: Seattle, USA, 2013.

35) "Neural connectivity of voice control using structural equation modeling". Flagmeier SG, Ray KL, Parkinson AL, Laird AR, Larson CR, Robin DA. Organization of Human Brain Mapping: Seattle, USA, 2013.

36) "Cognitive implications of component fractionation across BrainMap task-based ICA networks". Ray KL, Fox PM, McKay DR, Beckmann CF, Smith SM, Fox PT, Laird AR. Organization of Human Brain Mapping: Beijing, China, 2012.

37) "Converging hub identification: Evidence from resting state graph theory and BrainMap ICA analyses". Ray KL, Glahn DC, Blangero J, Laird AR. Organization of Human Brain Mapping: Beijing, China, 2012.

38) "Estimating BrainMap-based ICA dimensionality using cluster-based models". Ray KL, Fox PM, McKay DR, Beckmann CF, Smith SM, Fox PT, Laird AR. Organization of Human Brain Mapping: Quebec, Canada, 2011.

39) "Persistent and stable BrainMap-based ICA networks". McKay DR, Ray KL, Fox PM, Beckmann CF, Smith SM, Fox PT, Laird AR. Organization of Human Brain Mapping: Quebec, Canada, 2011.

40) "Differential functional connectivity of the medial and lateral orbitofrontal cortex". Zald DH, Ray KL, McHugo M, Eickoff SB, Glahn DC, Laird AR. Organization of Human Brain Mapping: Quebec, Canada, 2011.

41) "Functional assessment of intrinsic connectivity networks". Laird AR, Fox PM, Eickhoff SB, Ray KL, Turner JA, Glahn DC, Beckmann CF, Smith SM, Fox PT. Organization of Human Brain Mapping: Barcelona, Spain, 2010.

TEACHING AND MENTORSHIP

Teaching

2024	Functional MRI data analysis and its application in brainlife.io. INCF Training Days, UT Austin, 2024.
2024	"Resting State fMRI" I 320M: Topics in Health Informatics: Research Design and Analysis in Health Informatics, Guest Lecturer, UT Austin.
2023	"MACM for Network Analysis and Graph Theory Modelling", BrainMap Workshop, McGill University, Montreal, Canada.
2021	Introduction to Statistical Methods (PSY420M), Psychology, UT Austin Lab Instructor
2019	"Resting State: In search of a baseline" PSY 332D- Human Brain Imaging in Psychology, visiting lecturer, UT Austin

2019 "Meta-Analysis How To: Organization and Implementation"
Meta-Analysis and Functional Connectivity Workshop, University of New Hampshire, Durham, NH

2019 "MACM and Advanced Meta-Analytic Applications"
Meta-Analysis and Functional Connectivity Workshop, University of New Hampshire, Durham, NH

2019 "Functional Connectivity and Network Analysis using Graph Theory: Applications and Implementations"
Meta-Analysis and Functional Connectivity Workshop, University of New Hampshire, Durham, NH

2017 "Meta-Analysis How To: Organization and Implementation"
NeuroImaging Data Meta-Analysis Workshop, Shanghai Mental Health Center, Shanghai, China

2017 "MACM and Graph Theory Modeling: step by step"
NeuroImaging Data Meta-Analysis Workshop, Shanghai Mental Health Center, Shanghai, China

Mentorship and Student Training

Committee Member

Ariana White, Masters in Communication Sciences & Disorders, University of New Hampshire, 2024.
Lauren Keith, Masters in Communication Sciences & Disorders, University of New Hampshire, 2023.
Megan McMahon, Phd Cognitive Psychology, UT Austin, 2022.
Hannah Franz, Masters in Communication Sciences & Disorders, University of New Hampshire, 2019

Graduate Students

Stephen Kiilu, PhD Psychology, UT Austin, 2025 - current
Zhen Wei, Masters in Neuroscience, UT Austin, 2023 – 2024.
Esther Bennett, PhD Social Work, Rutgers University, 2022-2024
Megan McMahon, Phd Clinical Psychology, UT Austin, 2018 – 2022

Undergraduate Students

Alexandra Serrano, Texas A&M University, Summer 2026
Giselle Pedraza, UT Austin, 2026 Spring -
Anna Bostoganashvili, Centre College, Summer 2025
Karen Herrera, Vanderbilt University, Summer 2025
Sangmin (*Alden*) Yi, UT Austin, Spring 2024 – Summer 2025
Sohmee Kim, Neuroscience, UT Austin, 2024 – 2025.
Daniela Reyes, Psychology, UT Austin, 2024
Alisha Flexwala, UT Austin, 2023- 2026
Jedsada Thavornfung, Psychology, UT Austin, 2023 - 2025
Noah Haggerty, Biology, UT Austin, 2022 - 2022
Musa Raza, Biochemistry, UT Austin, 2021 - 2022
Rachel Kline, Psychology, St. Edwards University, 2022 - 2022
Grace Jumonville, Psychology & Plan II, UT Austin, 2021 – 2022
Javaria Baig, Neuroscience, UT Austin, 2020-2021
Angela Castillo, Neuroscience, UT Austin, 2020-2021

Sarah Meier, Communications Sciences & Disorders Senior Thesis, UNH, 2019
Noah Waller, Communications Sciences & Disorders Senior Thesis, UNH, 2019
Jackie Harle, Neuroscience, UT Austin, 2019-2021
Haana Janmohamed, Computer Science, UT Austin, 2018 – 2021

Staff

Melanie Collier, MS, Psychology, UT Austin, 2024 - present
Sangmin (Alden) Yi, Psychology, UT Austin, 2024 - 2025
Samantha Natal, Psychology, UT Austin, 2021 – 2023

AWARDS AND HONORS

Awards

2025 OHBM Caregiver Travel Award – OHBM, Brisbane AU.

2017 OHBM Merit Abstract Award – OHBM, Vancouver CA.

2014 Palmaz Academic Grant – Travel Award to OHBM 2014, University of Texas Health Science Center, San Antonio, Texas

2013 Palmaz Academic Grant – Travel Award to OHBM 2013, University of Texas Health Science Center, San Antonio, Texas

2012 Palmaz Academic Grant – Travel Award to OHBM 2012, University of Texas Health Science Center, San Antonio, Texas

2012 Presidential Student Ambassador Scholar, University of Texas Health Science Center, San Antonio, Texas

2011 Palmaz Academic Grant – Travel Award to OHBM 2011, University of Texas Health Science Center, San Antonio, Texas

2010–2014 Malcolm Jones Student Award, University of Texas Health Science Center, San Antonio, Texas

Research Grants

In review, being submitting or pending

Title: *An AI-Driven Meta-Analysis Platform for Investigating White Matter Connectivity in Human Neuroscience*
Source: *National Institutes of Health*
Lead Institution: The University of Texas
Requested Amount: $1,567,232
Dates: 04/01/2027-03/31/2032
PI/PD: Kimberly L Ray (Contact); Franco Pestilli
Co-PI(s): Angela Laird,

Title: BIDS 2.0+ ecosystem: AI-Ready Life-Science Data Standard.
Source: OS4LS – Open Science 4 Life Sciences
Lead Institution: Dartmouth
Requested Amount: $1,500,000
Dates: 12/01/2026-11/30/2027
PI- Y. Halchenko Co-PI: Kimberly L Ray

Title:	Developing community governance, infrastructure, and standards for brainlife.io
Source:	*National Science Foundation*
Lead Institution:	The University of Texas
Requested Amount:	$1,500,000
Dates:	In prep for Sept. 2026 submission
PI/PD:	Franco Pestilli
Co-PI(s):	Kimberly L Ray

INVITED PRESENTATIONS

2026 "BIDS Annual Open Science Opener"
OHBM Open Science, Bourdeaux, France

2025 "Distributed Infrastructures: Pros and Cons and Governance"
Distributed infrastructures for sharing and processing neuroimaging data, Brisbane, Australia

2025 "Data Governance Frameworks: BRIDGE-ing the gap between personal data privacy and open science."
OHBM Symposium, Brisbane, Australia

2025 "Coordinate based Neuroimaging Meta-Analysis"
OHBM Education Course, Brisbane, Australia

2024 "Data Governance within Neuroscience Community of Interest"
GA4GH – CONNECT, Melbourne, Australia

2024 "International Brain Data Governance Frameworks: facilitating global research"
2024 INCF Assembly, UT Austin

2024 "International Brain Data Governance Frameworks: developing a roadmap for facilitating global research" GA4GH Connect, Ascona, Switzerland.

2022 "Understanding brain network dysfunction in clinical populations" University of New Hampshire, Department of Biological Sciences, Durham, New Hampshire.

2020 "Reliability of resting state connectivity networks in individuals with elevated depression severity". Dallas Austin Area Memory Meeting, Zoom virtual meeting.

2018 "Organization and Interaction of Functional Brain Networks"
Psychology Department, Texas State University, San Marcos, TX

2018 "Functional Network Reorganization of Cognitive Control in Health and Schizophrenia"
Cognitive Neuroscience Seminar, University of Texas, Austin, TX

2018 "Functional Network Reorganization of Cognitive Control in Health and Schizophrenia"
Department of Psychology, Texas A&M University, College Station, Texas

2017 "Functional Network Reorganization of Cognitive Control in Health and Schizophrenia"
NeuroImaging and Psychology Seminar, Shanghai Mental Health Center, Shanghai, China

2017 "Meta-analysis and network analysis to study the functional organization of the human brain"
Department of Psychology, University of Texas, Austin, TX

2017 "Dynamic Reorganization of the Frontal Parietal Network during Cognitive Control and Episodic Memory" OHBM oral session

2016 "Functional Network Properties of Cognitive Control in Health and Schizophrenia"
Whistler Scientific Workshop on Brain Functional Organization, Connectivity and Behavior

2014 "Neuroimaging Meta-Analysis: ALE and BrainMap"
OHBM Educational Course (in place of Angela Laird)

2014 "A BrainMap approach to studying behavioral interpretations of intrinsic connectivity networks"
UC Davis, Sacramento, CA

2014 "Towards a prediction of cognitive deficits based on underlying connectivity differences in schizophrenia"
Whistler Scientific Workshop on Brain Functional Organization, connectivity and Behavior

2014 "A neuroinformatics approach to studying behavioral interpretations of intrinsic connectivity networks"
Florida International University, Miami, FL

2013 "The Connectome: Data mining and visualization"
OHBM Educational Course (*in place of Angela Laird*)

2013 "Delineating functional neural networks using the BrainMap database"
Texas Lutheran University – presented to the Physics Honor Society

ADMINISTRATIVE AND PROFESSIONAL SERVICE

Membership in professional societies

Society of Physics Students (SPS)
Organization for Human Brain Mapping (OHBM)
International Consortium of Neuroinformatics (INCF)
Society for Neuroscience (SfN)

Professional service & leadership

2023-current Leader, Data Governance Theme, Neuroscience Interest Group
(Partnership between BRIDGE + INCF + GA4GH)

2022-current Maintainer, Brain Imaging Data Structure (BIDS) Working Group

2015-2017 Panel Member, Imaging Research Center Graduate Advising Group
University of California, Davis

2013 Student Representative to the Radiological Science Committee on Graduate Studies (COGS)
University of Texas Health Science Center, San Antonio, Texas

Editorial Activities

Guest editor:

“Legal and ethical reforms for neurotechnological advancements in the global south: perspective and challenges.” Frontiers Research Topic.

Publication Coordination, NIH BRAIN CONNECTS Consortium:
Supported coordination of a consortium-wide Nature Portfolio publication package (~50 manuscripts) for the NIH BRAIN CONNECTS Initiative, assisting with publication tracking, cross-project communication, and milestone management.

Ad-Hoc Reviewer:
Scientific Reports, Frontiers in Neuroscience: Brain Imaging Methods, Frontiers in Human Neuroscience, Psychopharmacology, Human Brain Mapping, Psychiatry Research: Neuroimaging, NeuroImage, Schizophrenia Bulletin, Brain Imaging and Behavior, Biological Psychiatry, Physiology & Behavior, JAMA Psychiatry, Journal of Clinical Medicine, Biological Psychiatry: CNNI, Scientific Data, Brain and Behavior Functions.

Organization of scientific meetings

2026	*NIH BRAIN CONNECTS Annual Meeting, Washington DC, USA(Nov)*
2026	APEX BRAIN CONNECTS DCC Hackathon, Austin, TX, USA
2025	BRIDGE International Data Governance, Johannesburg, South Africa
2025	NIH BRAIN CONNECTS Annual Meeting, San Diego, CA, USA
2025	APEX BRAIN CONNECTS DCC Hackathon, Austin, TX, USA
2025	SOBP BrainHack Organizing Committee, Toronto, Canada
2024	SOBP BrainHack Organizing Committee, Austin, TX, USA
2024	INCF Assembly Program Committee, Austin, TX, USA
2024	BRIDGE International Data Governance, Austin, TX, USA